ITA0312 -MOBILE COMPUTING

EXPERIMENT-1

1. Use Figma to create an interactive diagram illustrating the components of a basic cellular system.

Aim:

To create an interactive diagram using Figma illustrating the major components of a basic cellular communication system and their functions.

Procedure:

1. Open Figma and create a new design file.
2. Create a frame and give it the title “Basic Cellular System”.
3. Add shapes representing the Mobile Station (MS), Base Transceiver Station (BTS), Base Station Controller (BSC), Mobile Switching Centre (MSC), and Public Switched Telephone Network (PSTN).
4. Arrange the components in a logical sequence from the Mobile Station to the PSTN.
5. Connect the components using arrows to represent the communication flow.
6. Add labels and short descriptions explaining the purpose of each component.
7. Use different colors and icons to distinguish the components clearly.
8. Create interactive elements using Figma's Prototype mode, such as clickable components that navigate to screens containing their descriptions.
9. Connect the prototype interactions and test them in Present mode.
10. Save the Figma design.

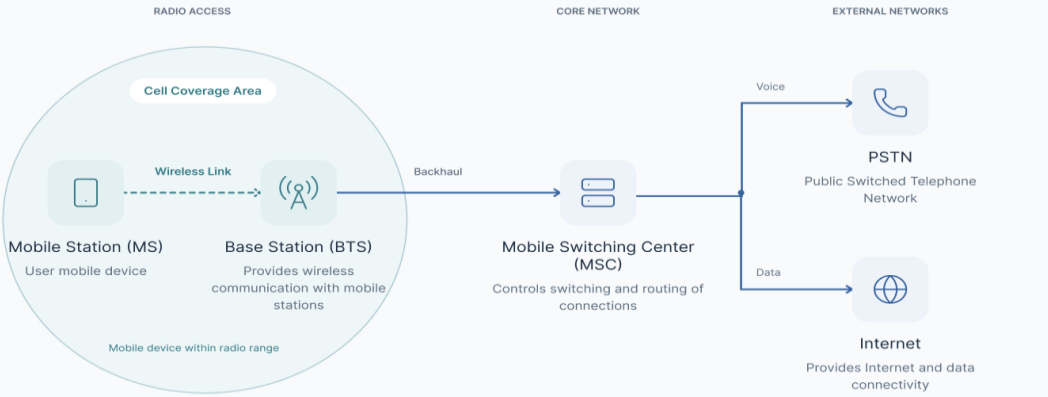
Basic cellular system

MOBILE COMPUTING LABORATORY

SYSTEM ARCHITECTURE · LAB RECORD

Use Figma to create an interactive diagram illustrating the components of a basic cellular system.

From a mobile device to voice and data networks: the essential components and communication path.



READING THE DIAGRAM

Communication flow · Cell coverage

MS → BTS → MSC → PSTN or Internet

The BTS provides radio access; the MSC routes connections onward to voice or data networks.

FIGURE 01 / Basic cellular system — conceptual architecture

Simplified for laboratory demonstration · Not to scale

▶ Start Communication

Cellular system components

MOBILE COMPUTING LABORATORY

SYSTEM ARCHITECTURE · LAB RECORD

CELLULAR SYSTEM COMPONENTS

The five essential components that connect a mobile device to voice and data networks.

RADIO ACCESS

Mobile Station (MS)

User mobile device that communicates with the cellular network.

RADIO ACCESS

Base Station (BTS)

Provides wireless communication between mobile stations and the network.

CORE NETWORK

Mobile Switching Center (MSC)

Controls switching and routing of connections.

EXTERNAL NETWORKS

PSTN

Public Switched Telephone Network for voice communication.

EXTERNAL NETWORKS

Internet

Provides Internet and data connectivity.

The BTS provides radio access; the MSC routes connections onward to voice or data networks.

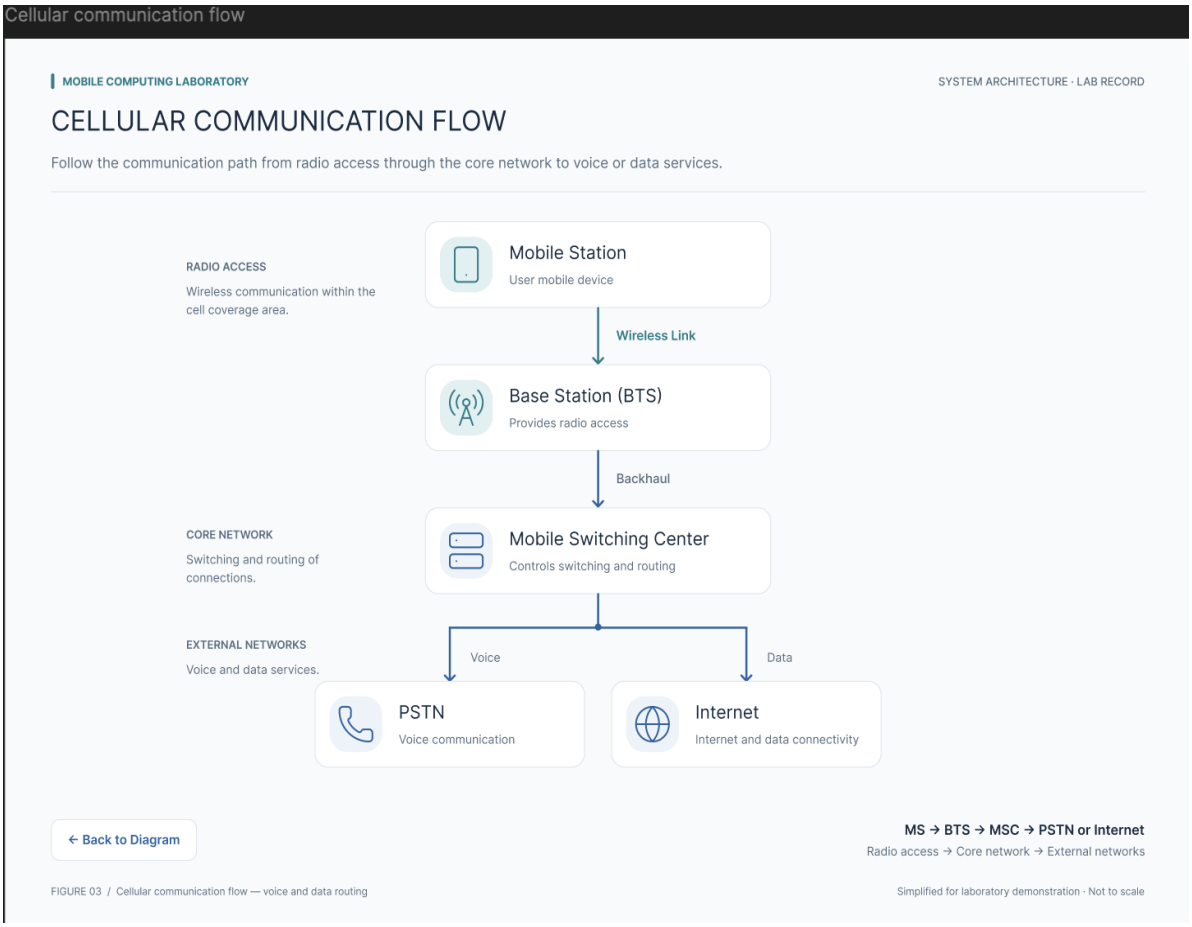
← Back to Diagram

MS → BTS → MSC → PSTN or Internet

Radio access → Core network → External networks

FIGURE 02 / Cellular system components — functions and roles

Simplified for laboratory demonstration · Not to scale



Result:

An interactive diagram illustrating the components of a basic cellular system and their roles in mobile communication was successfully designed using Figma.

EXPERIMENT-2

2. Design a user interface prototype for a mobile station using Figma, considering connectivity and convenience.

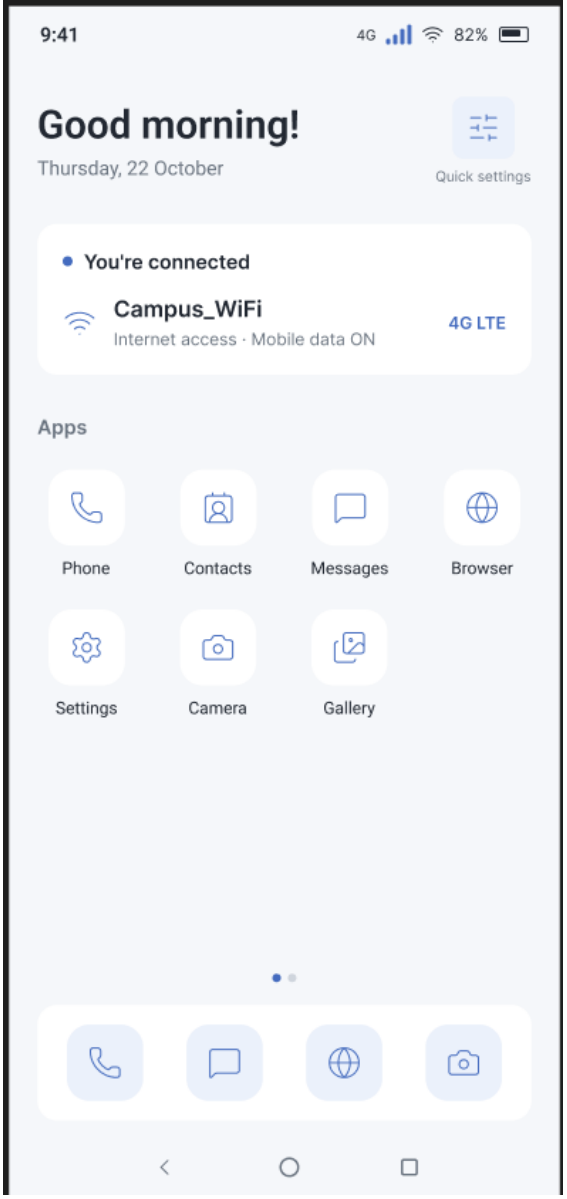
Aim:

To design a user interface prototype for a mobile station using Figma, considering connectivity, accessibility, and user convenience.

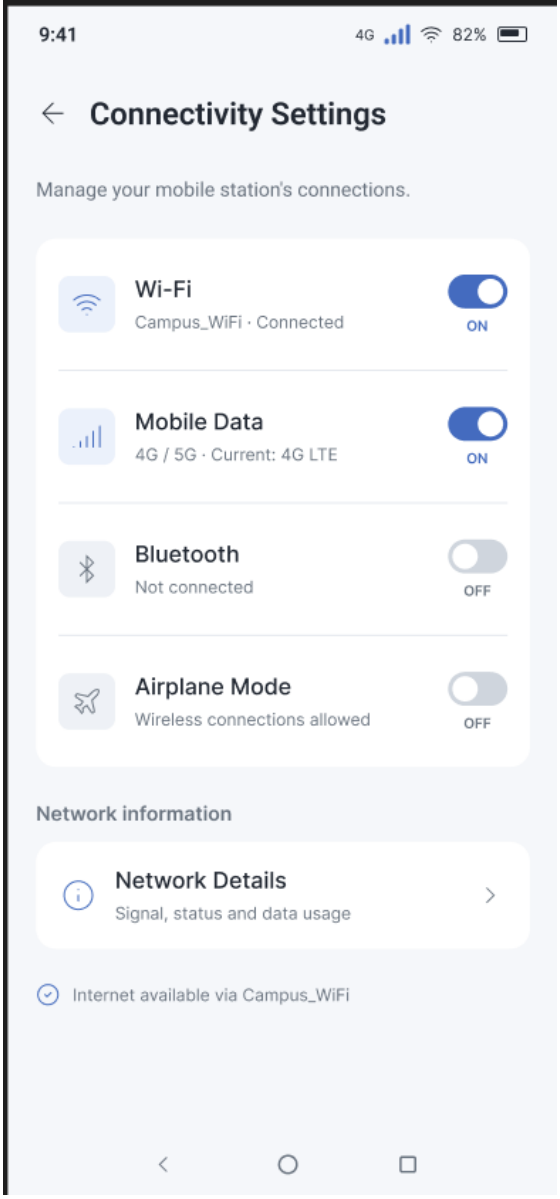
Procedure:

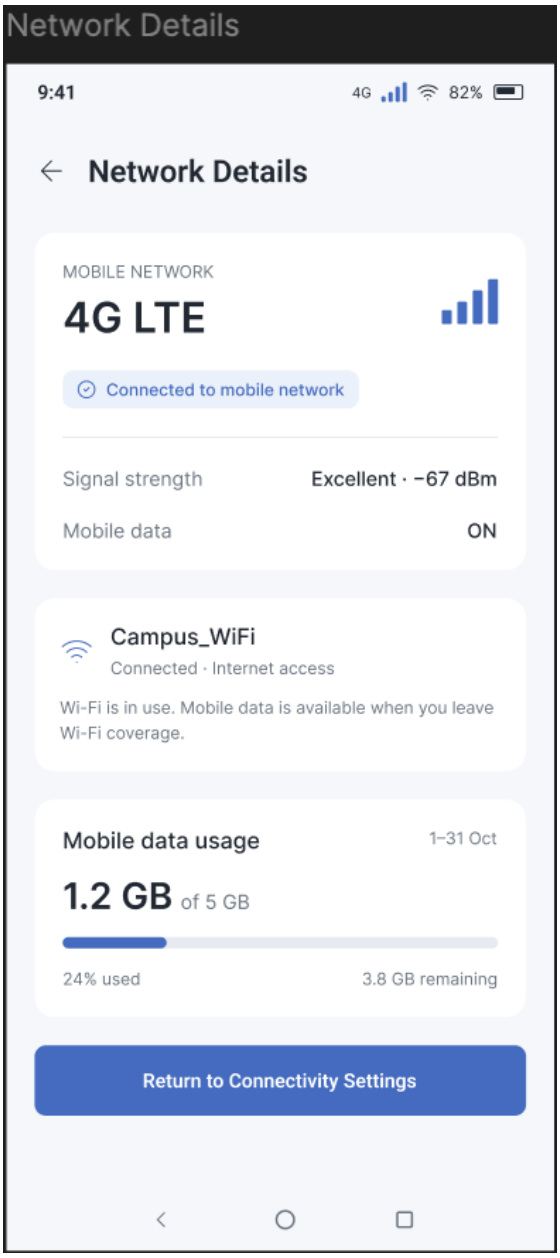
1. Open Figma and create a new design file.
2. Create a mobile-sized frame representing a smartphone screen.
3. Design a home screen containing essential features such as Phone, Contacts, Messages, Internet, and Settings.
4. Add a status bar displaying network signal strength, Wi-Fi connectivity, battery level, and time.
5. Design a connectivity section that allows users to access Wi-Fi, mobile data, Bluetooth, and airplane mode.
6. Use suitable icons, readable text, and a consistent color scheme to improve usability.
7. Add buttons and navigation elements for easy access to the main features.
8. Switch to Prototype mode and connect the buttons to relevant screens or interactive states.
9. Test the prototype in Present mode to verify navigation and connectivity controls.
10. Save the completed Figma prototype.

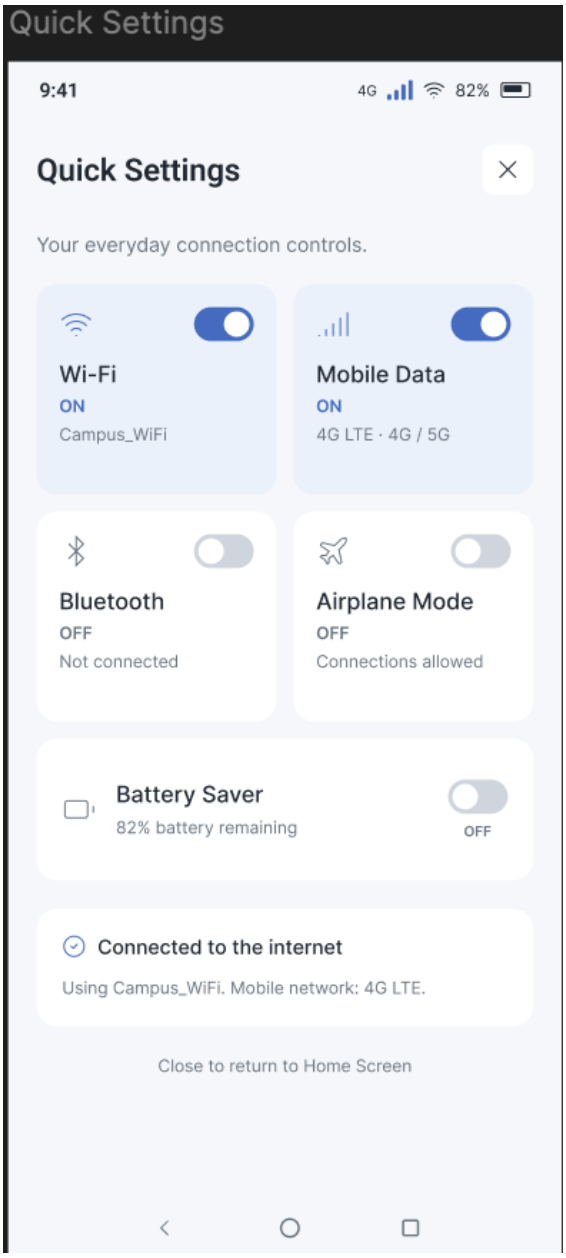
Home Screen



Connectivity Settings







Result:

A user-friendly mobile station interface prototype was successfully designed using Figma, incorporating connectivity controls, essential mobile features, and convenient navigation.

EXPERIMENT-3

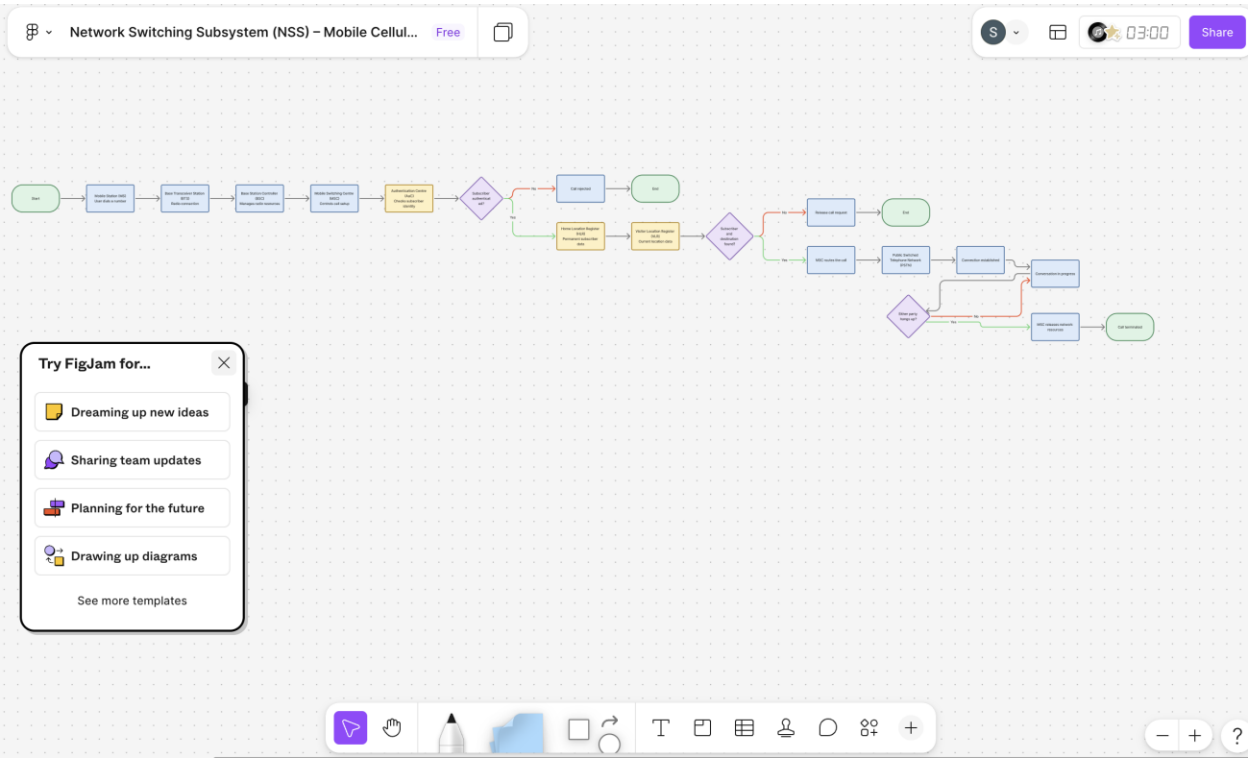
3. Develop a flowchart in Figma depicting the operation of a Network Switching Subsystem in a mobile cellular system.

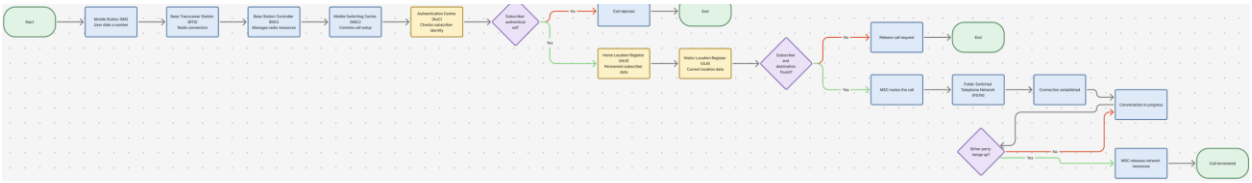
Aim:

To develop a flowchart using Figma depicting the operation of the Network Switching Subsystem (NSS) in a mobile cellular system, including subscriber verification, call routing, and connection establishment.

Procedure:

1. Open Figma and create a new design file.
2. Create a frame and add the title "Network Switching Subsystem in a Mobile Cellular System".
3. Add flowchart shapes representing the Mobile Station (MS), Base Transceiver Station (BTS), Base Station Controller (BSC), and Mobile Switching Centre (MSC).
4. Add the Home Location Register (HLR), Visitor Location Register (VLR), and Authentication Centre (AuC) to represent subscriber information and authentication.
5. Use arrows to connect the components and show the flow of communication.
6. Add decision diamonds for subscriber authentication and call-routing decisions.
7. Illustrate the steps of call initiation, subscriber verification, location lookup, call routing, connection establishment, and call termination.
8. Use different colors and labels to distinguish between the network components and operations.
9. Arrange the flowchart neatly and check that all components are connected correctly.
10. Save the design and use Present mode, if interactive elements are included, to demonstrate the flow.





Result:

A flowchart illustrating the operation of the Network Switching Subsystem was created using Figma.